








Advanced Time Series Analysis and Forecasting

Chapter 0: Refresher and inference for dependent data



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **ALFRED**: ArchivaL Federal Reserve Economic Data
- **AR**: AutoRegressive (model)
- **ARDL**: AutoRegressive Distributed Lag (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **ASDS**: Statistică aplicată și data science (the Master's programme Applied Statistics and Data Science)
- **ASE**: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- **ATS**: Advanced Time Series Analysis and Forecasting
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVAR**: Bayesian Vector AutoRegression
- **CBB**: Circular Block Bootstrap
- **CLT**: Central Limit Theorem
- **CPI**: Consumer Price Index
- **DGP**: Data-Generating Process
- **ECB**: European Central Bank
- **ECTS**: European Credit Transfer and Accumulation System
- **EOD**: End Of Day (daily closing data)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **EU**: European Union
- **EWC**: Equal-Weighted Cosine (estimator)
- **FDR**: False Discovery Rate
- **FM-OLS**: Fully Modified Ordinary Least Squares

Acronyms used in this material (2/2)

- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HICP**: Harmonised Index of Consumer Prices
- **INS**: Institutul Național de Statistică (the National Institute of Statistics (Romania))
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- **LLN**: Law of Large Numbers
- **LLSW**: Lazarus, Lewis, Stock and Watson (2018)
- **MA**: Moving Average
- **MBB**: Moving-Block Bootstrap
- **MDS**: Martingale Difference Sequence
- **MFM**: Modelling Financial Markets
- **MSE**: Mean Squared Error
- **NW**: Newey–West (standard errors)
- **NW-A**: Newey–West with the Andrews (1991) bandwidth
- **OLS**: Ordinary Least Squares
- **QR**: Quick Response (code)
- **QS**: Quadratic Spectral (kernel)
- **SARIMA**: Seasonal ARIMA (model)
- **SPA**: Superior Predictive Ability (test)
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Course overview

- **Programme**
  - Master's programme Applied Statistics and Data Science (ASDS), year 1, semester 2
  - Faculty of Cybernetics, Statistics and Economic Informatics, ASE Bucharest
  - 7 ECTS credits; 2 hours of lecture and 2 hours of seminar a week; academic year 2026/2027
- **Lecturer**
  - Prof. Daniel Traian Pele, danpele@ase.ro
- **Course website**
  - danpele.github.io/Advanced-Time-Series
  - slides, seminars, notebooks, Quantlets and the graded quizzes



The Bucharest University of Economic Studies (ASE)

Photo: Joe Mabel (2014); CC BY 3.0; Wikimedia Commons

Position in the curriculum

- **Before**: Time Series Analysis (TSA, bachelor), Chapters 0–10 of TSA
  - ARMA, ARIMA and unit roots, SARIMA, GARCH, VAR and Granger causality, cointegration and VECM, long memory, machine learning, state space models
  - assumed known: this course starts from them and does not repeat them
- **This course**: ATS, master, year 1, semester 2
  - methodology for forecasting, identification and inference with dependent data
  - macroeconomic (Romanian and EU), energy, climate and financial series
- **After**: Modelling Financial Markets (MFM, master, year 2, semester 1)
  - the same methods applied to market risk, portfolios, microstructure and digital assets

Evaluation

- **Team project and individual oral defence: 70%**
  - proposal and pre-registration (team): 5%
  - replication and extension: repository, report, AI_USE.md and AI_ERRORS.md (team): 15%
  - individual oral defence: 50%
- **Chapter quizzes: 20%**
  - online, on the course website, with the ASE Google account
- **Attendance: 10%**
  - recorded with the attendance form or the QR code
- There is no written exam

Textbooks

- **Main references**
  - Hamilton (1994): the theory of linear time series, unit roots and VAR
  - Kilian and Lütkepohl (2017): structural VAR analysis and identification
  - Durbin and Koopman (2012): state space methods
  - Petropoulos et al. (2022): forecasting theory and practice, open access
- **Bridge from the bachelor course**
  - Huang and Petukhina (2022); Hyndman and Athanasopoulos (2021)
- **Specialised**
  - Lütkepohl (2005); Shumway and Stoffer (2017); Baltagi (2021); Angelopoulos and Bates (2023); for this chapter also Lahiri (2003)
  - each chapter adds its landmark papers as case studies

Course map (1/2)

Ch.	Topic
0	Refresher and inference for dependent data
1	Forecast evaluation, scoring rules and combination
2	Structural breaks and nonlinear models
3	Structural VAR and local projections
4	Cointegration revisited: VECM, ARDL and panel data
5	Bayesian VAR, factor models and nowcasting
6	State space models and Bayesian filtering
7	Regime-switching models
8	Advanced volatility modelling: realised measures, HAR and multivariate GARCH

Course map (2/2)

Ch.	Topic
9	VaR, ES and backtesting: elicitability, scoring and model risk
10	Long memory and rough volatility
11	Spectral and wavelet analysis
12	Machine learning and deep learning for time series
13	Foundation models and conformal prediction
14	Causal inference for time series
15	Review and project defence
16	Explosive roots and bubbles (self-study)

Lectures and seminars

- **Lectures**
  - one chapter a week, built around landmark papers replicated on our data
  - each lecture ends with a section on AI for scientific discovery
- **Seminars**: each seminar comes *before* its lecture and opens with the notions it needs
  - Part A: derivations; Part B: estimation and inference on data, each exercise closing with an interpretation question; Part C: open problems that can seed the project
  - exercises marked [Solved], with full solutions, and exercises marked [Proposed], discussed in class
  - every seminar includes the critique of an answer given by an AI tool
- Seminars are practice and are not graded

The team project

- **Teams of up to 3 students**
  - replicate a landmark paper of the course, or another paper approved by the lecturer
  - extend it with the methods of the course, preferably on Romanian or EU data
- **Stages**
  - Stage 1: research question, data and an evaluation design fixed before any estimation (pre-registration)
  - Stage 2: replication, extension and a GitHub repository that reproduces every number (Colab)
  - Stage 3: presentation and individual oral defence
- Inference that is valid for dependent data (this chapter) is part of every project
- The project can be written and defended in Romanian or in English

AI policy

- **Allowed and declared**
  - AI_USE.md: tool, purpose, prompts, what was kept and what was changed
  - AI_ERRORS.md: at least three errors of AI tools caught by the team, and how each was detected
  - undeclared use counts as plagiarism
- **Checked**
  - every reference against its DOI, every number against the data and the code
- **Oral defence without AI**
  - each member explains the methods, the code and the results

Materials and tools

- **For every chapter**
  - lecture and seminar slides in English and Romanian
  - lecture and seminar notebooks in English, runnable in Google Colab
  - a Quantlet with the code of every chart, and a quiz
- **Data**
  - daily market data from EODHD (EOD Historical Data), saved in the course repository
  - public macroeconomic data: INS, BNR, Eurostat, the ECB and FRED
- **Software**
  - Python: statsmodels, arch, linearmodels, scikit-learn, PyTorch

Learning outcomes

- State the limit theorems behind every standard error used with time series: ergodic LLN, martingale-difference and mixing CLT
- Derive the long-run variance and its consequences for t-tests and confidence intervals
- Choose and defend a HAC estimator: kernel, bandwidth, critical values (standard or fixed-b)
- Use block bootstraps (moving, circular, stationary) and know when the wild bootstrap is not enough
- Measure size distortions by Monte Carlo and control data snooping
- Build a replication package that reproduces every number of a project

Data used in this chapter

Series	Definition	Period	Source
Romanian inflation	HICP, annual rate of change, %, monthly	January 2006 – August 2026	Eurostat
Romanian real GDP	chain-linked volumes, seasonally adjusted; growth q/q, %	2000Q1 – 2026Q2	Eurostat
EUR/RON	BNR reference rate; daily log change, %	5 July 2005 – 18 September 2026	BNR
S&P 500, BET	daily log returns and squared returns, %	2000 – 18 September 2026	EODHD
US macro	real GDP; 10-year and 3-month Treasury yields	1962 – 2026	FRED

- Daily series on their own trading calendars; Eurostat and FRED series read online at the build date
- HICP: harmonised index of consumer prices; the BNR target is defined on the national CPI, close to the HICP

Where the data come from



The National Bank of Romania (BNR), Bucharest: reference rates and the inflation target

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons



The Eccles Building, Washington: the Federal Reserve Board; FRED is run by the St. Louis Fed

Photo: Federal Reserve (2011); public domain; Wikimedia Commons

- A central bank publishes the series we test: the inflation target of the BNR is 2.5% $\pm$ 1 pp since 2013 (the AI mini-case at the end)

Notation (1/2)

- $\{x_t\}$, $t = 1, \ldots, T$: a weakly stationary series observed over T periods
  - $\mu = Ex_t$: its mean; $\gamma_j = \mathrm{Cov}(x_t, x_{t-j})$: the autocovariance at lag j, the same for every t
  - $\gamma_0 = \mathrm{Var}(x_t)$; $\rho_j = \gamma_j/\gamma_0 \in [-1, 1]$: the autocorrelation at lag j
- Sample counterparts (a hat marks an estimated value)
  - sample mean $\bar{x} = T^{-1} \sum_{t=1}^{T} x_t$
  - sample autocovariance $\hat{\gamma}_j = T^{-1} \sum_{t=j+1}^{T} (x_t - \bar{x})(x_{t-j} - \bar{x})$
  - $\hat{\gamma}_j$ averages the products of deviations from the mean that lie j periods apart
- Long-run variance $\Omega = \sum_{j=-\infty}^{\infty} \gamma_j = 2\pi f(0)$
  - the sum of all autocovariances ($\gamma_{-j} = \gamma_j$); it governs the precision of $\bar{x}$ (next section)
  - $f(\omega)$: the spectral density at frequency ω; $f(0)$ measures the variance of the slow, long-run movements

Notation (2/2)

- Kernel $k(\cdot)$: a weight function with $k(0) = 1$; lag j receives the weight $k(j/S)$
  - bandwidth S: the scale of the weights; lags much longer than S get (almost) zero weight
  - Newey–West: $L = S - 1$ lags for the Bartlett kernel; $b = S/T$: the bandwidth as a share of the sample
- Size of a test: the probability of rejecting a true H_0
  - nominal level 5% unless stated; a test over-rejects when its actual size exceeds the nominal level
- HAC: heteroskedasticity and autocorrelation consistent (an estimator of Ω)
  - HAR: heteroskedasticity and autocorrelation robust (the test built on a HAC estimator)

Prior knowledge from TSA (bachelor)

Topic	TSA	Used in ATS for
Stationarity, ACF, Wold	TSA, Chapter 1	long-run variance, HAC, bootstrap (this chapter)
ARMA	TSA, Chapter 2	benchmarks, Monte Carlo designs, sieve bootstrap
Unit roots, ARIMA	TSA, Chapter 3	nonstandard asymptotics, spurious regression, Chapters 2, 4, 16
GARCH	TSA, Chapter 5	martingale differences, Chapters 8–9
VAR, Granger	TSA, Chapter 6	structural VAR, local projections, BVAR (Chapters 3, 5)
Cointegration, VECM	TSA, Chapter 7	ARDL, panel cointegration (Chapter 4)
Long memory	TSA, Chapter 8	local Whittle, rough volatility (Chapter 10)
State space, Markov switching	TSA, Chapter 10	filtering and regimes (Chapters 6, 7)

- Slides of the bachelor course: TSA, Chapter 1 to TSA, Chapter 10; each ATS chapter links to the TSA chapter it extends

Stationarity and the Wold representation

- TSA, Chapter 1. Weak stationarity: constant mean and γ_j depending only on j
  - strict stationarity: all finite-dimensional distributions are invariant to a shift in time
  - neither implies the other without moment conditions: Cauchy i.i.d. is strictly but not weakly stationary (no variance)
- Wold theorem Hamilton (1994): every weakly stationary, purely nondeterministic series is a weighted sum of past innovations
  - $x_t - \mu = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}, \quad \psi_0 = 1, \quad \sum_j \psi_j^2 < \infty$
  - ε_t: white noise with variance σ^2, the one-step forecast error; ψ_j: the effect of a shock after j periods
  - ε_t is uncorrelated, not necessarily independent: GARCH innovations are Wold innovations
- Hence white noise is not enough for i.i.d.-based inference on squares or on extremes
  - In this chapter: $\Omega = \sigma^2 \big(\sum_j \psi_j \big)^2$; $\sum_j \psi_j$ is the cumulative (long-run) effect of a shock, so Ω is the variance of the permanent component

ARMA and unit roots (1/2)

- TSA, Chapter 2. ARMA(p, q): $\phi(L)x_t = \theta(L)\varepsilon_t$
  - L: the lag operator, $Lx_t = x_{t-1}$; $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$; $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$
  - stationary if the roots of $\phi(z)$ lie outside the unit circle; invertible if those of $\theta(z)$ do
- Long-run variance in closed form: $\Omega = \sigma^2 \theta(1)^2 / \phi(1)^2$
  - $\phi(1) = 1 - \sum_i \phi_i$ and $\theta(1) = 1 + \sum_i \theta_i$: the two polynomials evaluated at $z = 1$
  - AR(1): $\Omega = \sigma^2/(1-\phi)^2$; the closer $\phi(1)$ is to 0 (more persistence), the larger Ω

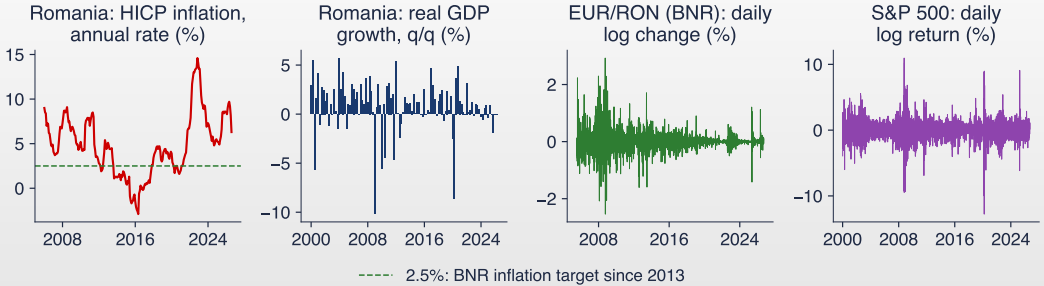
ARMA and unit roots (2/2)

- TSA, Chapter 3. Unit root, $\phi(1) = 0$: $\Omega = \infty$ and the mean is not defined
  - the t-statistic has a nonstandard (Dickey–Fuller) limit
- Spurious regression: one random walk regressed on an independent one
  - $|t| > 2$ in 76 of 100 regressions in Granger and Newbold (1974)
  - the t-statistic diverges at rate $\sqrt{T}$ Phillips (1986)
  - no HAC estimator repairs a spurious regression (Seminar 0, B5)
- Near unit roots (ϕ close to 1) are the hard case of this chapter
  - every HAC method over-rejects there (Monte Carlo section)

GARCH, VAR, cointegration, state space

- GARCH (TSA, Chapter 5): $r_t = \sigma_t z_t$ is a martingale difference; r_t^2 is an ARMA(1,1) with strong, slowly decaying autocorrelation
  - r_t: the return; σ_t: its conditional volatility; z_t: i.i.d. with mean 0 and variance 1
  - Means of returns: HAC barely matters; means of squared returns (variances, risk premia): HAC matters a lot
- VAR (TSA, Chapter 6): Granger causality tests are Wald tests; with heteroskedastic errors they need robust covariances
  - Local projections (Chapter 3) have serially correlated errors by construction: HAC is part of the method
- Cointegration (TSA, Chapter 7): long-run relations; the long-run variance reappears in FM-OLS and in the KPSS statistic
- State space (TSA, Chapter 10): the Kalman filter gives the Gaussian likelihood; its standard errors still need the sandwich form under misspecification

The series of this chapter



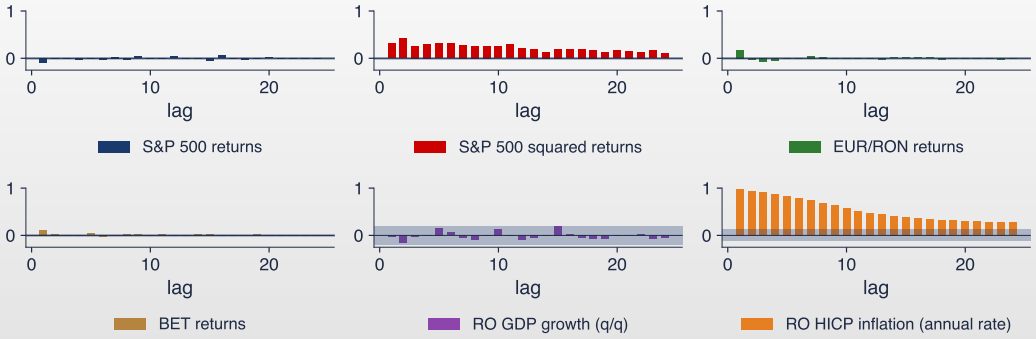
- Romanian HICP inflation, January 2006 – August 2026: from −2.9% (May 2016) to 14.6% (November 2022); last value 6.3%
- Romanian GDP growth: 106 quarters, minimum −10.2% in 2009Q1; daily: 5 352 EUR/RON and 6 714 S&P 500 observations

ATS_ch0_dependence_data

Interpreting the four series

- Inflation moves in long swings: years above and years below the 2.5% line
  - Each monthly value shares eleven months with the previous one (annual rate): overlap adds to persistence
- GDP growth looks close to white noise, apart from two crisis quarters
  - With $T = 106$, two outliers dominate the sample variance
- EUR/RON and the S&P 500: volatility clusters (calm years, crisis bursts): dependence in the second moment
- What do you think? Which of the four series has the largest long-run variance relative to its variance?
  - Answer: inflation, the most persistent series; the ratios are on the slide "Interpreting the autocorrelations"

Autocorrelation in six series



- Sample ACF, lags 1–24, with the $\pm 1.96/\sqrt{T}$ band of the i.i.d. hypothesis (shaded)
- $\hat{\rho}_1$: S&P 500 -0.10, squared 0.31, EUR/RON 0.17, BET 0.11, GDP -0.03, inflation 0.98

ATS_ch0_dependence_data

Interpreting the autocorrelations

- S&P 500 returns: small negative $\hat{\rho}_1$; squared returns: all 24 lags above the band, $\hat{\rho}_{12} = 0.22$
  - Long-run variance over variance (Andrews bandwidth): 0.8 for returns, 5.7 for squared returns
- BET and EUR/RON: positive $\hat{\rho}_1$ (thin trading, central-bank smoothing of the reference rate)
  - Ratios 1.2 and 1.1: modest, but enough to move a borderline t-statistic
- Inflation: $\hat{\rho}_1 = 0.98$ and a ratio of 27.0: a naive standard error is about five times too small
- The rest of the chapter: what these ratios mean, how to estimate them, and when the estimates fail

Recap: the refresher map

- TSA gave the models; ATS needs the inference that goes with dependent data
- White noise is not independence: GARCH returns are uncorrelated, their squares are not
- The long-run variance $\Omega = \sigma^2\theta(1)^2/\phi(1)^2$ links ARMA models to the precision of a sample mean
- In our data the dependence ranges from negligible (GDP growth) to extreme (inflation, squared returns)

The variance of a mean under dependence

- Exact, for any weakly stationary series: the variance of the scaled mean adds up the covariances of all pairs of observations
  - $\mathrm{Var}(\sqrt{T}\,\bar{x}) = \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T}\gamma_{t-s} = \sum_{|j|<T}\left(1-\frac{|j|}{T}\right)\gamma_j$
  - γ_{t-s}: the covariance of x_t and x_s; $T-|j|$ pairs lie at distance j, hence the weight $1-|j|/T$
- If $\sum_j |\gamma_j| < \infty$, the weights tend to 1 (dominated convergence):
  - $\mathrm{Var}(\sqrt{T}\,\bar{x}) \to \Omega = \gamma_0 + 2\sum_{j\geq 1}\gamma_j = 2\pi f(0)$
- The i.i.d. formula uses γ_0 only; the error factor is $\Omega/\gamma_0 = 1 + 2\sum_{j\geq 1}\rho_j$
  - factor above 1: the naive standard error is too small; below 1: too large
  - effective sample size $T\gamma_0/\Omega$: the number of independent observations with the same precision
  - AR(1) with $\phi = 0.9$: 1000 observations are worth about 53 independent ones
- Two questions for the rest of the section
  - LLN: when does $\bar{x} \to \mu$?
  - CLT: when is $\sqrt{T}(\bar{x}-\mu)/\sqrt{\Omega}$ asymptotically $N(0,1)$?

Ergodicity

- A strictly stationary $\{x_t\}$ is **ergodic** if every shift-invariant event (unchanged when the whole path is shifted in time) has probability 0 or 1
  - Intuition: one long path visits the whole distribution; time averages equal ensemble averages
- **Ergodic theorem** Birkhoff (1931): stationary, ergodic, $E|x_t| < \infty \Rightarrow \bar{x} \to Ex_t$ almost surely
  - Functions of finitely many lags of an ergodic process are ergodic: sample moments, autocovariances, OLS
- Counterexample: $x_t = Z + \varepsilon_t$, with Z a random level drawn once and ε_t i.i.d.: stationary, not ergodic, $\bar{x} \to Z$
  - A single history of one economy is a single draw of Z: why regime and break questions matter (Chapters 2, 7)



George David Birkhoff (1884–1944)

Photo: unknown author (c. 1910); public domain; Wikimedia Commons

Mixing conditions (1/2)

- Strong (α-) mixing coefficient Rosenblatt (1956):
  - $\alpha(m) = \sup_t \sup\{|P(A \cap B) - P(A)P(B)| : A \in \mathcal{F}^t_{-\infty}, B \in \mathcal{F}^\infty_{t+m}\}$
  - $\mathcal{F}^t_{-\infty}$: the events determined by x_s, $s \leq t$ (the past); $\mathcal{F}^\infty_{t+m}$: those determined by x_s, $s \geq t + m$ (the future after a gap of m periods)
  - $|P(A \cap B) - P(A)P(B)|$ is 0 when A and B are independent: $\alpha(m)$ is the largest dependence between a past and a future event m periods apart
- α-mixing: $\alpha(m) \to 0$ as $m \to \infty$, i.e. the distant past and future become independent
  - $\beta(m) = \sup_t E\big[\sup_{B \in \mathcal{F}^\infty_{t+m}} |P(B \mid \mathcal{F}^t_{-\infty}) - P(B)|\big]$: the average gap between conditional and unconditional probabilities of future events
  - $\varphi(m) = \sup_t \sup\{|P(B \mid A) - P(B)| : A \in \mathcal{F}^t_{-\infty}, P(A) > 0, B \in \mathcal{F}^\infty_{t+m}\}$: the worst case over past events
  - $\alpha(m) \leq \beta(m) \leq \varphi(m)$: β- and φ-mixing are stronger; mixing implies ergodicity Bradley (2005)

Mixing conditions (2/2)

- Models that are mixing
  - stationary ARMA with continuous innovations: geometrically β-mixing Mokkadem (1988)
  - strictly stationary GARCH(1,1), stochastic volatility: geometrically β-mixing Carrasco and Chen (2002)
  - geometric: $\beta(m) \leq C\lambda^m$ for some constants $C > 0$ and $0 < \lambda < 1$
- Models that are not
  - unit roots, long memory with $d > 0$ (TSA, Chapter 8), some deterministic chaos
  - there mixing fails, or is too slow for the standard CLT
- Mixing coefficients cannot be estimated from one path
  - the assumption is checked through the model, not tested

Laws of large numbers for dependent data

- L^2 **law**: weakly stationary with $\gamma_j \to 0 \Rightarrow E(\bar{x} - \mu)^2 \to 0$
  - Proof: $E(\bar{x} - \mu)^2 = T^{-1} \sum_{|j|<T} (1 - |j|/T)\gamma_j$, and the Cesàro mean (the average of the first terms) of the sequence $\gamma_j \to 0$ tends to 0
- **Ergodic law** Birkhoff (1931): almost sure convergence, only $E|x_t| < \infty$, no second moment
  - Covers squared returns of a GARCH with infinite fourth moment: means converge, but the CLT may fail
- **Mixingales and near-epoch dependence** Hamilton (1994)
  - mixingale: $\|E(x_t - \mu \mid \mathcal{F}_{t-m})\|_2 \leq c_t \psi_m$ with $\psi_m \to 0$: the past predicts less and less of x_t
  - near-epoch dependence: $\|x_t - E(x_t \mid v_{t-m}, \ldots, v_{t+m})\|_2 \to 0$ as $m \to \infty$, with $\{v_t\}$ mixing: x_t is almost a function of nearby values of a mixing process
  - $\|z\|_2 = (Ez^2)^{1/2}$; c_t: scaling constants; these cover functions of mixing processes, e.g. the squares of a GARCH
- Consistency (LLN) is cheap; valid standard errors (CLT plus a consistent $\hat{\Omega}$) are the hard part

Martingale differences and their central limit theorem

- $\{u_t\}$ is a **martingale difference sequence** (MDS) if $E(u_t \mid \mathcal{F}_{t-1}) = 0$
  - $\mathcal{F}_{t-1}$: the information available at $t-1$; the past does not predict u_t on average
  - Uncorrelated with every function of the past, but may be conditionally heteroskedastic: GARCH returns, scores of a correct likelihood, one-step forecast errors of an optimal forecast
- **MDS CLT** Billingsley (1961): stationary, ergodic MDS with $Eu_t^2 = \sigma^2 < \infty \Rightarrow \sqrt{T}\,\bar{u} \to_d N(0, \sigma^2)$
  - $\to_d$: convergence in distribution; nonstationary MDS: Lindeberg-type conditions Brown (1971)
  - Here $\Omega = \gamma_0$: no autocorrelation correction is needed, only heteroskedasticity-robust errors White (1980)
- Overlapping h-step forecast errors are not an MDS: they follow an MA$(h-1)$ (Section "Overlapping observations")

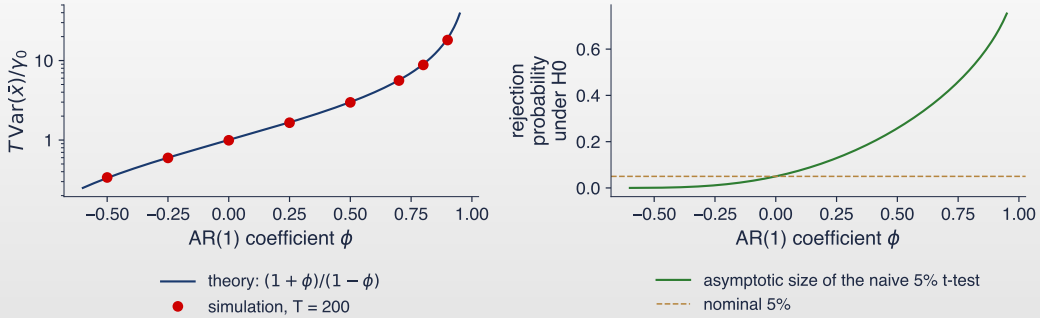
Central limit theorems for mixing and linear processes (1/2)

- **Mixing CLT** Ibragimov (1962): if x_t is stationary and
  - $E|x_t|^{2+\delta} < \infty, \quad \sum_m \alpha(m)^{\delta/(2+\delta)} < \infty, \quad \Omega > 0 \Rightarrow \sqrt{T}(\bar{x} - \mu) \to_d N(0, \Omega)$
  - $\delta > 0$: the number of moments beyond the second; $\alpha(m)$: the strong mixing coefficient defined above
  - trade-off: more moments (larger δ) allow slower mixing; the first CLT under strong mixing is Rosenblatt (1956)
- **Linear processes** Phillips and Solo (1992): $x_t - \mu = \sum \psi_j \varepsilon_{t-j}$ with i.i.d. or MDS ε_t and $\sum j|\psi_j| < \infty$
  - the condition on ψ_j requires the effect of a shock to die out fast enough

Central limit theorems for mixing and linear processes (2/2)

- Beveridge–Nelson decomposition of the lag polynomial $\psi(L) = \sum_j \psi_j L^j$:
  - $\psi(L) = \psi(1) - (1-L)\tilde{\psi}(L)$, so $\sqrt{T}(\bar{x} - \mu) = \psi(1)\sqrt{T}\bar{\varepsilon} + o_p(1)$
  - $\psi(1) = \sum_j \psi_j$: the long-run effect of a shock; $\tilde{\psi}(L)$: the polynomial of the transitory part; $o_p(1)$: a term that tends to 0 in probability
  - the mean of x_t behaves like $\psi(1)$ times the mean of the innovations, hence $\Omega = \sigma^2\psi(1)^2$
- Functional versions (invariance principles) give the Brownian-motion limits
  - they underlie the unit-root theory and the fixed-b theory (next section)
- All three CLTs need $0 < \Omega < \infty$
  - unit roots give $\Omega = \infty$; over-differencing gives $\Omega = 0$

Information lost to dependence



⊡ Left: $T\mathrm{Var}(\bar{x})/\gamma_0$ for an AR(1), theory $(1+\phi)/(1-\phi)$ and 20 000 simulations with $T = 200$; right: asymptotic size of the naive 5% test, $2[1 - \Phi(1.96/\sqrt{(1+\phi)/(1-\phi)})]$, with Φ the standard normal distribution function

ATS_ch0_long_run_variance

Interpreting the variance inflation

- $\phi = 0.5$: the factor is 3 (simulated 2.98); $\phi = 0.9$: 19 (simulated 18.1, the finite-T factor $1 - |j|/T$ lowers it)
  - The naive test rejects a true H_0 with probability 15% at $\phi = 0.3$, 26% at 0.5 and 65% at 0.9
- Negative dependence works the other way: $\phi = -0.5$ gives a factor 0.33, a conservative naive test
  - The S&P 500 mean return is an example: its HAC standard error is smaller than the naive one
- Confidence intervals inherit the factor: a naive 95% interval covers μ only about 74% of the time at $\phi = 0.5$

Regression with dependent errors (1/2)

- Model $y_t = x_t'\beta + u_t$ with $E(x_t u_t) = 0$
  - x_t: the vector of regressors; β: the coefficients; u_t: the error, uncorrelated with x_t; $'$ denotes transposition
- Under a CLT for the products $x_t u_t$, the OLS estimator is asymptotically normal:
  - $\sqrt{T}(\hat{\beta} - \beta) \to_d N(0,\ Q^{-1}\Omega_{xu}Q^{-1}), \quad Q = E(x_t x_t'), \quad \Omega_{xu} = \sum_j E(x_t u_t u_{t-j} x_{t-j}')$
  - Q: the second-moment matrix of the regressors; Ω_{xu}: the long-run variance of $x_t u_t$
  - the form $Q^{-1}\Omega_{xu}Q^{-1}$ is called a sandwich: Ω_{xu} between two copies of Q^{-1}

Regression with dependent errors (2/2)

- Scalar slope, x_t and u_t independent stationary series: the error factor is
  - $\Omega_{xu}/(\gamma_{x,0}\gamma_{u,0}) = 1 + 2\sum_{j\geq 1}\rho_x(j)\rho_u(j)$
  - $\gamma_{x,0}, \gamma_{u,0}$: the variances of x_t and u_t; $\rho_x(j), \rho_u(j)$: their autocorrelations at lag j
- A large distortion needs both a persistent regressor and a persistent error
  - if either autocorrelation is zero, the factor is 1; a persistent predictor with persistent errors is the worst case
- The Newey–West covariance matrix Newey and West (1987): the sandwich $\hat{Q}^{-1}\hat{\Omega}_{xu}\hat{Q}^{-1}$
  - $\hat{Q}$: the sample mean of $x_t x_t'$; $\hat{\Omega}_{xu}$: a kernel estimate of Ω_{xu} (next section)

Recap: asymptotics for dependent data

- $\mathrm{Var}(\sqrt{T}\bar{x}) \to \Omega = \sum_j \gamma_j = 2\pi f(0)$; the i.i.d. formula keeps only γ_0
- Ergodicity gives the LLN; mixing or martingale structure plus moments give the CLT
- For an MDS, $\Omega = \gamma_0$: heteroskedasticity-robust errors suffice
- Positive persistence makes naive tests over-reject; negative persistence makes them conservative

Estimating the long-run variance

- Plugging in all sample autocovariances fails: $\sum_{|j|<T} \hat{\gamma}_j = 0$ identically for demeaned data
  - high-order $\hat{\gamma}_j$ use few pairs and are mostly noise
- **Kernel estimator**: a weighted sum of sample autocovariances
  - $\hat{\Omega} = \sum_{|j|<T} k(j/S)\,\hat{\gamma}_j, \quad k(0) = 1, \quad k(x)$ decreasing in $|x|$
  - equivalently, a smoothed periodogram at frequency 0: $\hat{\Omega} = 2\pi \hat{f}(0)$ Shumway and Stoffer (2017)
- Consistency: $S \to \infty$ and $S/T \to 0$, under mixing and moment conditions Andrews (1991)
  - the down-weighted lags create bias, the included ones create variance: S balances the two
- A variance estimator must be nonnegative: not every kernel guarantees this

Case study: Newey and West (1987) (1/2)

- Newey and West (1987), Econometrica 55(3): the Bartlett kernel $k(x) = 1 - |x|$ for $|x| \leq 1$, with L lags
  - $\hat{\Omega}_{NW} = \hat{\Gamma}_0 + \sum_{j=1}^{L} \left(1 - \frac{j}{L+1}\right)(\hat{\Gamma}_j + \hat{\Gamma}_j')$, $\quad \hat{\Gamma}_j = T^{-1} \sum_{t>j} \hat{h}_t \hat{h}_{t-j}'$
- Notation
  - $\hat{h}_t = x_t \hat{u}_t$: regressor times OLS residual $\hat{u}_t$ (the score of observation t)
  - $\hat{\Gamma}_j$: the sample autocovariance matrix of $\hat{h}_t$ at lag j; $\hat{\Gamma}_0$: its variance matrix
  - $1 - j/(L+1)$: weights that decrease linearly from 1 to $1/(L+1)$; lags beyond L get weight 0

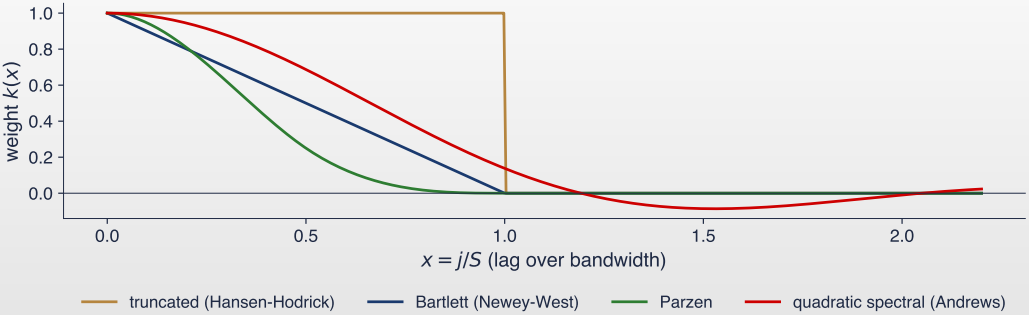
Case study: Newey and West (1987) (2/2)

- **Positive semi-definite by construction**: Bartlett weights are the autocovariances of a moving sum
  - up to end effects, $\hat{\Omega}_{NW} = \frac{1}{(L+1)T} \sum_t \left(\sum_{i=0}^{L} \hat{h}_{t-i} \right) \left(\sum_{i=0}^{L} \hat{h}_{t-i} \right)'$
  - a sum of outer products vv', each positive semi-definite, so the sum is too
  - spectral view: the Bartlett (Fejér) window is nonnegative at every frequency
- Consistency with $L \to \infty$, $L = o(T^{1/4})$ in the original paper
  - $L = o(T^{1/4})$: L grows more slowly than $T^{1/4}$; Andrews (1991) refines the rates
- One page of algebra, now the default robust standard error in every econometrics package

A truncated kernel can give a negative variance

- Truncated (uniform) kernel: $k(x) = 1$ for $|x| \le 1$: $\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{L} \hat{\gamma}_j$ (used by Hansen and Hodrick (1980) for overlapping forecasts)
  - Exactly right when $\gamma_j = 0$ for $j > L$, e.g. an MA($h-1$) forecast error with $L = h-1$
- Ten observations alternating in sign, $(1, -1, 1, -1, 1, -1, 1, -1, 2, -2)$: $\hat{\gamma}_0 = 1.6$, $\hat{\gamma}_1 = -1.3$, $\hat{\gamma}_2 = 1.0$
  - Truncated, $L = 1$: $\hat{\gamma}_0 + 2\hat{\gamma}_1 = -1.0 < 0$
  - Bartlett, $L = 1$: $\hat{\gamma}_0 + \hat{\gamma}_1 = 0.3 > 0$
- Lesson: a negative variance is not a numerical accident; it is a property of the kernel (Seminar 0, A3–A4)

Kernels



- Weight $k(x)$ against $x = j/S$, the lag relative to the bandwidth; Bartlett as in the Newey–West case study
- Parzen: $1 - 6x^2 + 6|x|^3$ ($|x| \le 1/2$), $2(1 - |x|)^3$ ($1/2 < |x| \le 1$); QS (quadratic spectral): $\frac{25}{12\pi^2 x^2}\left[\frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5)\right]$, all lags

ATS_ch0_long_run_variance

Interpreting the kernels

- Bartlett, Parzen and QS have nonnegative spectral windows (the Fourier transform of the weights): $\hat{\Omega} \geq 0$ always
  - QS takes small negative weights (minimum -0.086) yet stays positive semi-definite
- Kernel order q: $1 - k(x) \sim c|x|^q$ near 0, with $c > 0$ a constant; Bartlett $q = 1$, Parzen and QS $q = 2$
  - bias of order S^{-q}, variance of order S/T: the optimal S grows like $T^{1/(2q+1)}$
  - i.e. $T^{1/3}$ for Bartlett and $T^{1/5}$ for QS
- Andrews (1991): QS minimises the asymptotic MSE among kernels with nonnegative estimates

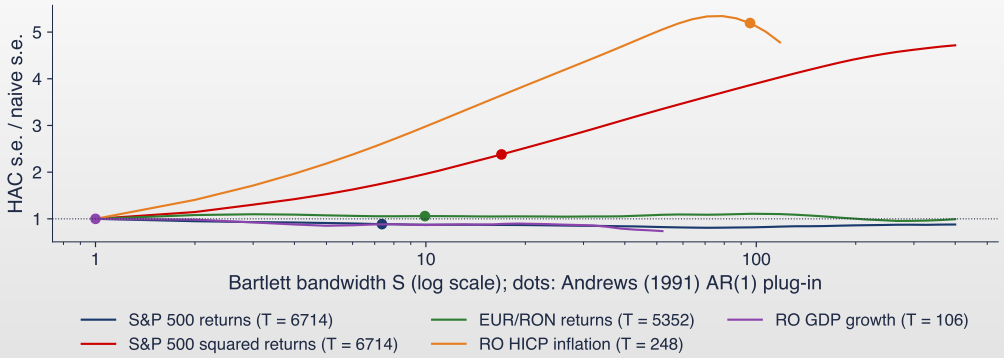
Case study: Andrews (1991) and the choice of bandwidth (1/2)

- MSE-optimal bandwidth Andrews (1991): the S that minimises the mean squared error of $\hat{\Omega}$
  - $S^* = c_k \left(\alpha(q)\, T\right)^{1/(2q+1)}$
  - q: the order of the kernel; c_k: a constant of the kernel; $\alpha(q)$: a function of the unknown spectrum, large for persistent series
  - Bartlett: $S^* = 1.1447(\hat{\alpha}(1)\, T)^{1/3}$; Parzen: $2.6614(\hat{\alpha}(2)\, T)^{1/5}$; QS: $1.3221(\hat{\alpha}(2)\, T)^{1/5}$
- **AR(1) plug-in**: fit an AR(1) with coefficient $\hat{\rho}$ to $\hat{h}_t$, then
  - $\hat{\alpha}(1) = \dfrac{4\hat{\rho}^2}{(1-\hat{\rho})^2(1+\hat{\rho})^2}, \qquad \hat{\alpha}(2) = \dfrac{4\hat{\rho}^2}{(1-\hat{\rho})^4}$
  - both grow without bound as $\hat{\rho} \to 1$: more persistence, wider bandwidth

Case study: Andrews (1991) and the choice of bandwidth (2/2)

- Newey–West rule of thumb: $L = \lfloor 4(T/100)^{2/9} \rfloor$
  - $\lfloor \cdot \rfloor$: the integer part; L depends only on T, not on the data
  - their nonparametric automatic selection Newey and West (1994) estimates the spectrum instead
- Prewhitening with a VAR(1), then recolouring Andrews and Monahan (1992)
  - filter out the AR(1) part, estimate Ω on the residuals, then undo the filter: less bias for persistent series
- The MSE criterion targets $\hat{\Omega}$, not the test
  - this is the source of the size distortions of the next sections

HAC standard errors as a function of the bandwidth



- Ratio of the Bartlett HAC standard error of the mean to the naive one, S from 1 to min(400, $T/2$); dots: Andrews AR(1) plug-in

ATS_ch0_hac

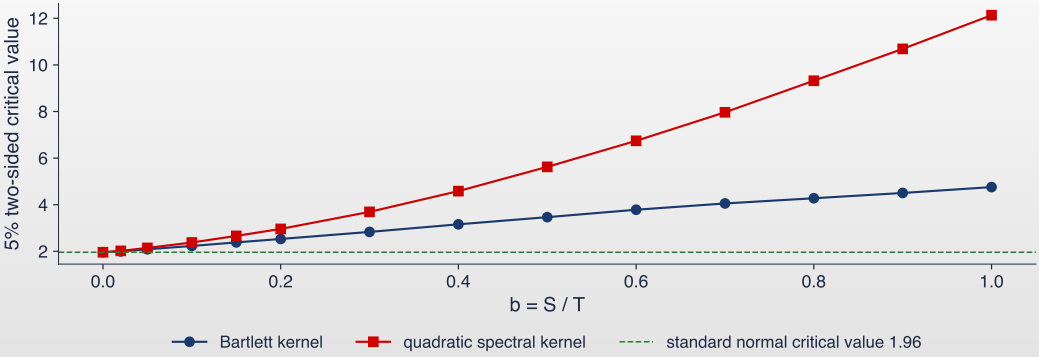
Interpreting the bandwidth paths

- Squared S&P 500 returns: ratio 2.04 with the rule of thumb ($S = 11$), 2.38 with Andrews ($S = 17$), still rising to 4.72 at $S = 400$
  - No plateau: the AR(1) plug-in misses the slow decay of volatility autocorrelation (long memory, Chapter 10)
- Romanian inflation: 2.18 with the rule of thumb, 5.19 with Andrews ($S = 96$ months)
  - The rule of thumb ignores the data: with T fixed it gives the same S for white noise and for a near unit root
- S&P 500 returns: ratio below 1 at every S (negative $\hat{\rho}_1$); EUR/RON: flat near 1.06

Fixed-b asymptotics

- Standard theory: $S/T \to 0$, $\hat{\Omega} \to_p \Omega$, the t-statistic is $N(0,1)$
  - $\to_p$: convergence in probability; the sampling error of $\hat{\Omega}$ is ignored
  - with 100–300 observations, $\hat{\Omega}$ is noisy and biased downwards: the normal critical values are too small
- **Fixed-b** Kiefer, Vogelsang and Bunzel (2000), Kiefer and Vogelsang (2002), Kiefer and Vogelsang (2005): keep $b = S/T$ fixed as $T \to \infty$
  - $\hat{\Omega}/\Omega \to_d Q_k(b), \qquad t \to_d W(1)/\sqrt{Q_k(b)}$
  - $W(r)$: a standard Brownian motion on $r \in [0,1]$ (r: the fraction of the sample); $Q_k(b)$: a random functional of the Brownian bridge $\tilde{W}(r) = W(r) - rW(1)$
  - Bartlett kernel: $Q(b) = \frac{2}{b}\int_0^1 \tilde{W}(r)^2\,dr - \frac{2}{b}\int_0^{1-b} \tilde{W}(r+b)\tilde{W}(r)\,dr$ Kiefer and Vogelsang (2005)
  - nonstandard but pivotal (free of unknown parameters): critical values depend only on the kernel and on b
- Higher-order theory: fixed-b critical values remove the leading size distortion of the normal approximation Sun, Phillips and Jin (2008)

Fixed-b critical values



- Two-sided 5% critical values of the t-test for a mean with $S = bT$, simulated (20 000 i.i.d. normal samples, $T = 500$)

ATS_ch0_hac

Interpreting the fixed-b critical values

- Bartlett: 2.23 at $b = 0.1$, 3.47 at $b = 0.5$, 4.76 at $b = 1$ (Kiefer–Vogelsang–Bunzel statistic)
  - Close to the cubic approximation reported in Kiefer and Vogelsang (2005) for the Bartlett kernel
- QS: 2.38 at $b = 0.1$ and 5.62 at $b = 0.5$: QS with a large b is very noisy
- Larger b: less bias in $\hat{\Omega}$, more variance, compensated by a larger critical value: better size, lower power
- What do you think? Why does the critical value at $b \to 0$ tend to 1.96?
  - Answer: as $b \to 0$, $Q_k(b) \to 1$ ($\hat{\Omega}$ becomes consistent), so $t \to_d W(1) \sim N(0,1)$: the standard asymptotics return

Recommendations for practice: Lazarus, Lewis, Stock and Watson (2018) (1/2)

- Lazarus et al. (2018) choose the bandwidth for the *test*: a size–power frontier, not the MSE of $\hat{\Omega}$
  - first recommendation: Newey–West with $S = 1.3\sqrt{T}$ and fixed-b critical values
- Second recommendation: the equal-weighted cosine (EWC) estimator
  - $\hat{\Omega} = \nu^{-1} \sum_{j=1}^{\nu} \hat{\Lambda}_j^2, \qquad \hat{\Lambda}_j = \sqrt{2/T} \sum_t \cos[\pi j (t - 1/2)/T]\, \hat{h}_t, \qquad \nu = 0.4 T^{2/3}$
  - $\hat{\Lambda}_j$: the projection of $\hat{h}_t$ on the j-th cosine, a low-frequency component; each $\hat{\Lambda}_j^2$ estimates Ω
  - ν: the number of cosines averaged, also the degrees of freedom: critical values from Student t_ν

Recommendations for practice: Lazarus, Lewis, Stock and Watson (2018) (2/2)

- The size–power trade-off is unavoidable
  - Lazarus, Lewis and Stock (2021) give the frontier for kernel tests
- Very persistent series (ϕ near 1): no kernel or EWC rule is reliable
  - Müller (2014) proposes tests valid under local-to-unity dependence (a root that tends to 1 as T grows)
- These are the defaults of the course from Chapter 1 on

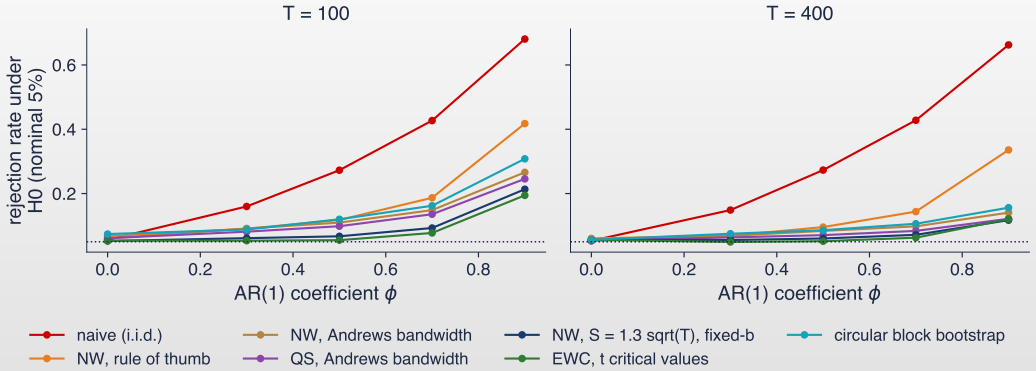
Recap: HAC estimation

- Kernel estimators down-weight distant lags; Bartlett (Newey–West), Parzen and QS are always nonnegative; the truncated kernel is not
- The Andrews (1991) bandwidths are MSE-optimal for $\hat{\Omega}$ and adapt to persistence; the rule of thumb does not
- Fixed-b critical values account for the noise in $\hat{\Omega}$; LLSW: NW with $S = 1.3\sqrt{T}$ or EWC with $\nu = 0.4\,T^{2/3}$

Monte Carlo design

- DGP: $x_t = \phi x_{t-1} + \varepsilon_t$, $\varepsilon_t \sim N(0,1)$ i.i.d., 200 burn-in values; $H_0 : \mu = 0$ is true
  - ϕ: the AR(1) coefficient, i.e. the persistence; burn-in: initial values dropped so that the series starts from its stationary distribution
  - $\phi \in \{0, 0.3, 0.5, 0.7, 0.9\}$, $T \in \{100, 400\}$; 5 000 replications (1 000 for the bootstrap, 399 resamples each); seed 2026
- Seven tests at nominal 5%
  - naive i.i.d.; NW rule of thumb; NW and QS with the Andrews AR(1) bandwidth (normal critical values)
  - NW with $S = 1.3\sqrt{T}$ and fixed-b values; EWC with t_ν; circular block bootstrap with the Politis–White block length
- Monte Carlo standard error of a rejection rate p estimated from R replications: $\sqrt{p(1-p)/R}$
  - about 0.3 percentage points (pp) at $p = 5\%$ and $R = 5000$
- The same design as the Monte Carlo sections of Andrews (1991) and Kiefer and Vogelsang (2005): AR(1) data, rejection rates under the null

Size of seven tests under AR(1) dependence



- Rejection rate of $H_0 : \mu = 0$ (true) against ϕ; dotted line: 5%

ATS_ch0_size_monte_carlo

Size in numbers (%)

	$T = 100$			$T = 400$		
Test	$\phi = 0$	0.5	0.9	$\phi = 0$	0.5	0.9
naive i.i.d.	5.6	27.3	68.1	5.3	27.3	66.2
NW, rule of thumb	6.8	11.8	41.8	6.1	9.6	33.6
NW, Andrews	6.0	11.0	26.6	5.3	8.4	14.1
QS, Andrews	6.1	9.8	24.6	5.4	7.1	12.1
NW, $1.3\sqrt{T}$, fixed-b	5.2	6.7	21.3	5.6	6.0	11.7
EWC, t_ν	5.3	5.5	19.4	5.6	5.2	12.1
circular block bootstrap	7.4	12.0	30.8	5.6	8.6	15.6

- Monte Carlo standard errors: about 0.7 pp near 30%; bootstrap rows: about twice as large

Interpreting the Monte Carlo

- Naive test: 27.3% at $\phi = 0.5$ and 68.1% at $\phi = 0.9$; more data do not help (66.2% at $T = 400$)
  - The distortion is a property of the estimator, not of the sample size
- NW with the rule of thumb: 41.8% at $\phi = 0.9$, $T = 100$; Andrews bandwidths halve the excess; fixed-b and EWC come closest (21.3%, 19.4%)
  - At $T = 400$ and $\phi = 0.5$: 6.0% and 5.2%, essentially exact
- The block bootstrap is not a free lunch: 30.8% at $\phi = 0.9$, $T = 100$, between NW and fixed-b
- Our inflation series has $\hat{\rho}_1 = 0.98$: outside the range where any of these tests is reliable (AI mini-case)

Recap: size distortions

- Always report the actual size of the procedure you use, at the persistence of your data
- Fixed-b NW and EWC (Lazarus et al. (2018)) dominate the classical choices in size at a modest cost in power
- A Monte Carlo is an experiment: fix the seed, report R and the Monte Carlo standard error

The failure of the i.i.d. bootstrap

- Efron's bootstrap Efron (1979) resamples single observations: the dependence is destroyed
  - a star marks a bootstrap quantity: $\bar{x}^*$ is the mean of one resample, Var^* the variance over resamples
  - $\mathrm{Var}^*(\sqrt{T}\bar{x}^*) = \hat{\gamma}_0$, not Ω: the i.i.d. bootstrap reproduces the naive standard error
- Singh (1981): even for m-dependent data (independent beyond distance m) the i.i.d. bootstrap of the mean is inconsistent
  - Idea of the block bootstrap: resample blocks long enough to keep the dependence inside them
- Price to pay: dependence across block joints is lost, and the block length becomes a tuning parameter



Bradley Efron (2007)

Photo: Ryan K. Morris, NSTMF (2007); public domain; Wikimedia Commons

Case study: the moving-block bootstrap of Künsch (1989)

- Künsch (1989): overlapping blocks $B_i = (x_i, \ldots, x_{i+l-1})$ of length l, $i = 1, \ldots, T - l + 1$
  - draw $k = \lceil T/l \rceil$ blocks with replacement ($\lceil \cdot \rceil$: rounding up), concatenate, keep the first T values
- Consistency for smooth functions of means if $l \to \infty$, $l/T \to 0$
  - $\mathrm{Var}^*(\sqrt{T}\bar{x}^*) \approx \sum_{|j|<l}(1 - |j|/l)\hat{\gamma}_j$: a Bartlett estimate with $S = l$
  - Optimal $l \propto T^{1/3}$ for variance and bias, as for the Bartlett bandwidth Lahiri (2003)
- Edge effect: end observations enter fewer blocks, so $E^*\bar{x}^* \neq \bar{x}$
  - E^*: the expectation over resamples; centre the bootstrap statistics at $E^*\bar{x}^*$



Hans Rudolf Künsch (2007)

Photo: Renate Schmid (2007); CC BY-SA 2.0 de; Wikimedia Commons

Circular and stationary bootstrap

- **Circular block bootstrap** (Politis and Romano (1992); Lahiri (2003)): wrap the data on a circle, $x_{T+i} = x_i$
  - Every observation enters l blocks: $E^*\bar{x}^* = \bar{x}$ exactly (Seminar 0, A5–A6)
- **Stationary bootstrap** Politis and Romano (1994): block lengths geometric with mean $1/p$, uniform starting points, data wrapped on a circle
  - Equivalently: each step continues the block with probability $1 - p$ or jumps to a random point with probability p
  - The resampled series is stationary; the method is less sensitive to the choice of $1/p$ than MBB to l
- Efficiency: MBB and CBB have a smaller asymptotic MSE for the variance than the stationary bootstrap at their optimal lengths Lahiri (2003)
- The stationary bootstrap is the engine of the Reality Check White (2000) and of the SPA test (Chapter 1)

Choosing the block length (1/2)

- Politis and White (2004), corrected by Patton, Politis and White (2009): plug-in estimate of the MSE-optimal mean block length
  - $\hat{b}_{opt} = \left(\frac{2\hat{G}^2}{\hat{D}}\right)^{1/3} T^{1/3}, \quad \hat{G} = \sum_{|k|\leq M} \lambda(k/M)\,|k|\,\hat{\gamma}_k, \quad \hat{g} = \sum_{|k|\leq M} \lambda(k/M)\,\hat{\gamma}_k$
  - $\hat{D}_{SB} = 2\hat{g}^2$ (stationary), $\hat{D}_{CB} = \frac{4}{3}\hat{g}^2$ (circular)
- Notation
  - $\hat{G}$: autocovariances weighted by their lag, the bias term; $\hat{g}$: an estimate of Ω; $\hat{D}$: the variance term
  - λ: the flat-top (trapezoid) kernel; $M = 2\hat{m}$: the truncation lag
  - $\hat{m}$: the first lag after which K_T consecutive autocorrelations are insignificant

Choosing the block length (2/2)

- The same structure as the Andrews bandwidth
  - bias term G over variance term D, rate $T^{1/3}$: persistent series get longer blocks
- Implemented in the Python package `arch` (`optimal_block_length`)
- Our data
  - about 145 days for squared S&P 500 returns, 25 months for inflation
  - 0.8 for GDP growth, i.e. the i.i.d. bootstrap

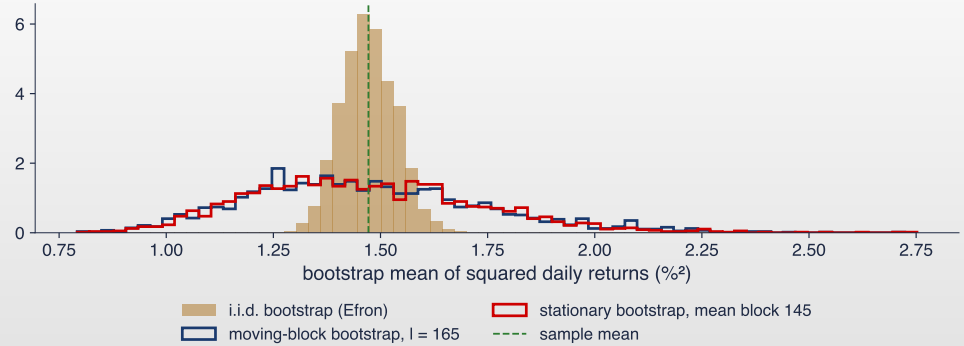
The wild bootstrap and its limits (1/2)

- **Wild bootstrap** Wu (1986), Liu (1988), Mammen (1993): each residual is multiplied by a random sign-like weight
  - $y_t^* = x_t'\hat{\beta} + \hat{u}_t\eta_t, \quad \eta_t$ i.i.d., $E\eta_t = 0$, $E\eta_t^2 = 1$
  - y_t^*: the bootstrap observation; η_t: Rademacher (± 1 with probability $1/2$ each) or the Mammen two-point distribution
- It keeps the heteroskedasticity of each $\hat{u}_t$ and destroys every autocorrelation
  - valid when the scores are an MDS: AR models with GARCH-type errors Gonçalves and Kilian (2004)
  - **not** valid for overlapping forecasts, local projections or the mean of squared returns

The wild bootstrap and its limits (2/2)

- Dependent wild bootstrap Shao (2010): η_t drawn from a stationary process
  - $\mathrm{Corr}(\eta_t, \eta_s) = a(|t-s|/l)$, with $a(\cdot)$ a kernel and l a bandwidth
  - wild in form, block-like in effect: neighbouring residuals keep their joint signs
- Sieve bootstrap Bühlmann (1997): fit an AR(p) with $p \to \infty$, resample its residuals
  - efficient for linear processes, fragile for nonlinear ones

Three bootstraps of one mean



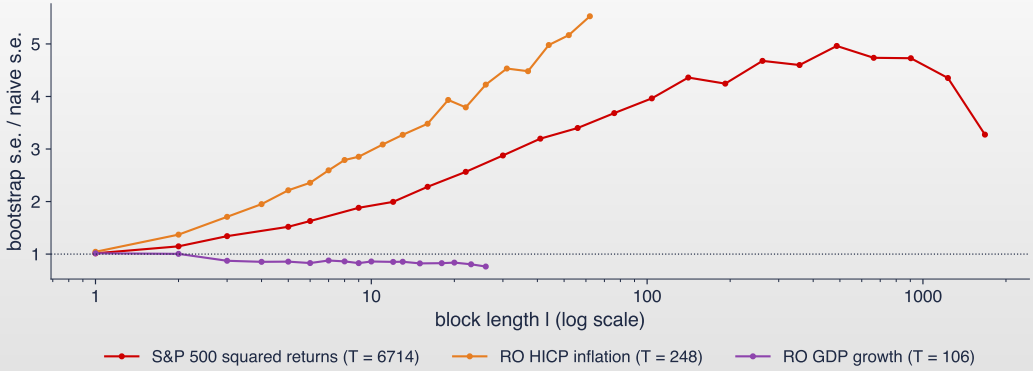
⊡ Mean of squared daily S&P 500 returns (1.472, $T = 6\,714$); 1999 resamples each; block lengths from Politis–White

ATS_ch0_block_bootstrap

Interpreting the three bootstraps

- i.i.d. bootstrap: standard error 0.063, equal to the naive 0.064, as theory says
  - Block bootstraps: MBB 0.276, CBB 0.268, stationary 0.269: about 4.2 times the naive value
- HAC with the Andrews bandwidth: 0.152 (about 2.4 times the naive value): smaller than the block bootstraps
  - Block length 145 against an Andrews bandwidth of 17: two plug-in rules, two different views of the same slowly decaying ACF
- When two valid methods disagree by a factor of two, report both and explain why: here, the volatility of volatility has long memory

Bootstrap standard error against the block length



⊡ Circular block bootstrap, 999 resamples per length; ratio to the naive standard error

ATS_ch0_block_bootstrap

Interpreting the block-length paths

- $l = 1$ is the i.i.d. bootstrap (ratio 1); the ratio grows with l while l is shorter than the memory of the series
  - Inflation: ratio 4.2 at the Politis–White length of 25 months
- GDP growth: flat near 1 (and slightly below for long blocks): no dependence to capture
- Very long blocks: few distinct blocks, noisy and downward-biased variance (the bias of $\hat{\gamma}_j$ at large lags)
- What do you think? Why does the inflation curve keep rising until l is about a quarter of the sample?
  - Answer: the autocorrelation of inflation decays very slowly (annual rates overlap, $\hat{\rho}_1 = 0.98$); only very long blocks capture it before the downward bias takes over

Means of six series: standard errors by method

Series	T	Mean	$\hat{\rho}_1$	naive	NW-A	stat. boot.	t_{naive} / t_{EWC}
S&P 500 returns	6 714	0.025	−0.10	0.0148	0.0131	0.0129	1.67 / 2.07
S&P 500 squared	6 714	1.472	0.31	0.064	0.152	0.269	23.06 / 6.71
BET returns	6 670	0.064	0.11	0.0171	0.0188	0.0187	3.73 / 2.89
EUR/RON	5 352	0.007	0.17	0.0038	0.0040	0.0040	1.86 / 1.73
RO GDP growth	106	0.829	−0.03	0.240	0.240	0.240	3.46 / 3.67
RO inflation	248	4.799	0.98	0.218	1.131	0.957	22.05 / 5.92

- NW-A: Newey–West with the Andrews bandwidth; stationary bootstrap with the Politis–White mean block; returns and changes in % per day, GDP in % per quarter, inflation in %

Interpreting the table

- S&P 500: t rises from 1.67 (naive) to 2.03 (fixed-b, critical value 1.99) and 2.07 (EWC): the mean return (6.2% a year) becomes borderline significant
  - Negative autocorrelation shrinks the standard error: robust inference is not always more conservative
- EUR/RON: mean depreciation of the leu not significant at 5% by any method (t between 1.73 and 1.86)
- Inflation: t falls from 22.05 to 5.92 for $H_0 : \mu = 0$; standard errors grow by a factor of about 4–5
- GDP growth: all methods agree; the question there is the two crisis quarters, not the dependence (Seminar 0, B2)

Recap: the bootstrap for dependent data

- The i.i.d. bootstrap reproduces the naive standard error; resample blocks instead
- MBB Künsch (1989) is a Bartlett HAC estimator in disguise; CBB removes the edge bias; the stationary bootstrap keeps stationarity
- Block length $\propto T^{1/3}$, chosen by Politis–White; the wild bootstrap handles heteroskedasticity only

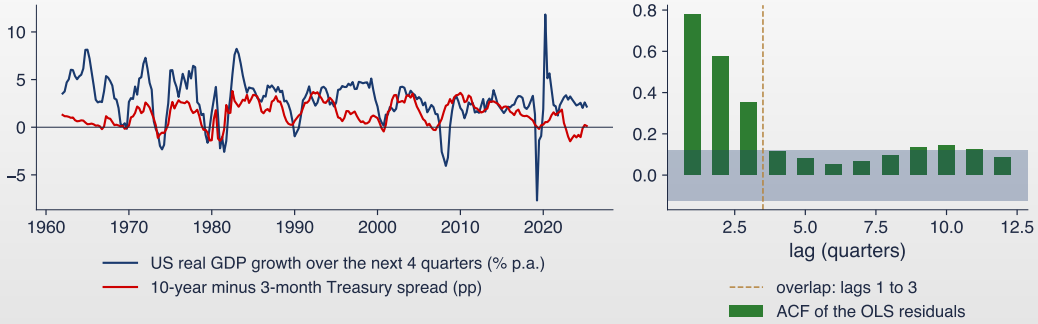
Case study: Estrella and Hardouvelis (1991) (1/2)

- Estrella and Hardouvelis (1991): does the slope of the yield curve predict US real growth?
  - regression of the average growth over the next h quarters on the term spread
  - $y_t^{(h)} = \frac{400}{h} \ln \frac{\mathrm{GDP}_{t+h}}{\mathrm{GDP}_t} = \beta_0 + \beta_1 (i_t^{10y} - i_t^{3m}) + u_t, \quad h = 4$
- Notation
  - $y_t^{(h)}$: real GDP growth from quarter t to $t+h$, annualised, in %; the factor $400/h$ turns a log change over h quarters into % a year
  - i_t^{10y}, i_t^{3m}: the 10-year and 3-month Treasury yields; their difference is the term spread
  - $\beta_1 > 0$: a steeper curve announces faster growth; u_t: the forecast error

Case study: Estrella and Hardouvelis (1991) (2/2)

- Overlap: $y_t^{(4)}$ and $y_{t+1}^{(4)}$ share three quarters
  - so u_t is at least MA(3) under H_0, even if quarterly shocks are independent
  - classical remedy: Hansen and Hodrick (1980) (truncated kernel, $h-1$ lags) or Newey–West with $L = h-1$
- Our data: FRED real GDP, 10-year and 3-month Treasury yields (quarterly averages)
  - 1962Q1 – 2025Q2, $T = 254$
- Same design for local projections (Chapter 3) and for long-horizon return regressions (MFM)

The spread and future growth



- Left: the two series; right: ACF of the OLS residuals, $\hat{\rho}_1 = 0.78$, $\hat{\rho}_3 = 0.35$, $\hat{\rho}_4 = 0.11$

ATS_ch0_term_spread

Six estimators of the standard error of the slope

Standard error	$\widehat{se}(\hat{\beta}_1)$	t	Critical value
Classical OLS	0.11	4.72	1.96
White (heteroskedasticity only)	0.10	–	1.96
Hansen–Hodrick, 3 lags	0.20	–	1.96
Newey–West, 3 lags	0.17	3.10	1.96
NW, Andrews ($S = 14$)	0.23	2.25	1.96
NW, $S = 1.3\sqrt{T} = 21$, fixed-b	0.25	2.04	2.18

- $\hat{\beta}_1 = 0.52$ pp of annual growth per pp of spread, $R^2 = 0.08$

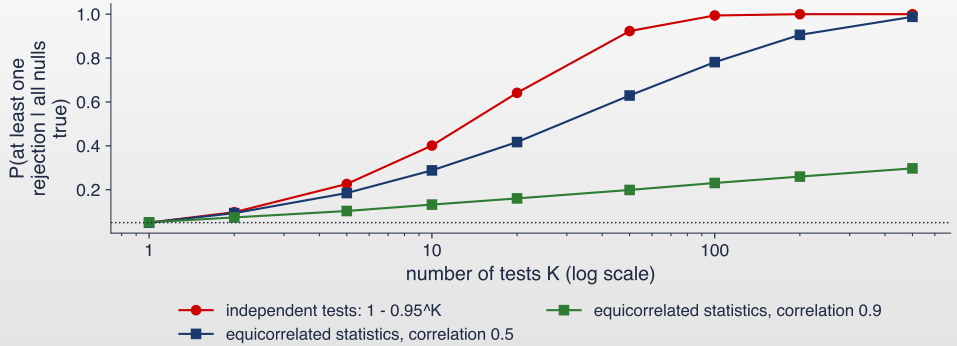
Interpreting the term-spread regression

- The residuals are more persistent than the MA(3) implied by overlap ($\hat{\rho}_4 = 0.11$): 3 lags are not enough
  - The standard error doubles from classical OLS to fixed-b: t from 4.72 to 2.04, just below the critical value 2.18
- Subsamples (NW, 3 lags): 1962–1988 $\hat{\beta}_1 = 1.12$ ($t = 4.6$); 1989–2025 $\hat{\beta}_1 = 0.15$ ($t = 0.9$)
  - The predictive power of the spread weakened after the original sample: a break question (Chapter 2) and an out-of-sample question (Chapter 1)
- Robust inference first, then stability: a full-sample t of 4.72 hides both problems

Data snooping

- **Data snooping**: reusing one data set to select a model, a rule or a specification, then testing it on the same data as if it were chosen in advance
  - Lo and MacKinlay (1990): sorting portfolios on characteristics already known to be related to returns biases asset-pricing tests
  - Harvey, Liu and Zhu (2016): more than 300 published return factors; a new factor needs $t > 3$, not $t > 2$
- Error rates for K tests
  - FWER (family-wise error rate): probability of at least one false rejection among the K tests at level α
  - Bonferroni tests each hypothesis at α/K; Holm (step-down; Holm (1979)) compares the ordered p-values $p_{(i)}$ with $\alpha/(K - i + 1)$; both control the FWER
  - FDR (false discovery rate): expected share of false rejections among rejections, controlled by Benjamini and Hochberg (1995)
- In forecasting: many models, many horizons, many samples
  - the honest p-value is the one of the *search*, not of the winner

The best of K tests



- Probability that the largest $|t|$ of K tests exceeds 1.96 when all nulls are true: independent and equicorrelated statistics (20 000 simulations)

ATS_ch0_data_snooping

Interpreting the family-wise error

- Independent tests: FWER 64% with $K = 20$ and 99.4% with $K = 100$
  - Bonferroni would test each at $0.05/20 = 0.0025$
- Correlated statistics (similar rules, overlapping samples): FWER 42% ($K = 20$) and 78% ($K = 100$) at correlation 0.5; 23% at 0.9
  - Bonferroni ignores the correlation and is conservative: the bootstrap of the maximum (next slide) uses it
- What do you think? With 50 moving-average rules, how many "significant" rules do you expect by chance at 5%?
  - Answer: $50 \times 0.05 = 2.5$ if the rules were independent; correlated rules give fewer distinct rejections, but the best t is still inflated

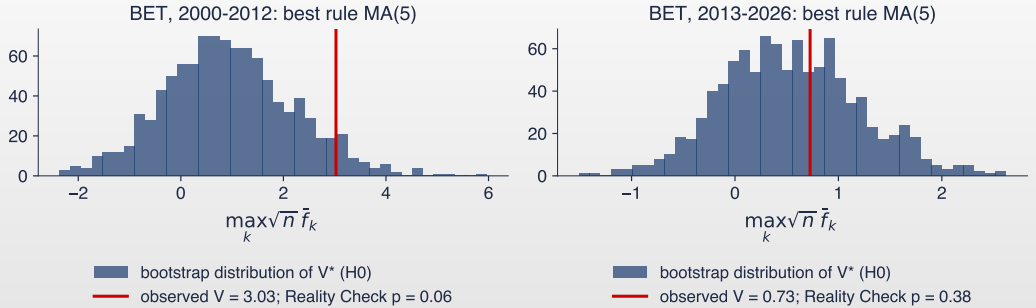
Case study: the Reality Check of White (2000) (1/2)

- K models against a benchmark, evaluated on n periods White (2000)
  - $f_{k,t}$: the performance difference of model k at t (e.g. return of rule k minus buy-and-hold); $\bar{f}_k$: its mean over the n periods
- Hypothesis and statistic
  - $H_0 : \max_k Ef_{k,t} \leq 0$ (no model beats the benchmark); $\quad \bar{V} = \max_k \sqrt{n}\,\bar{f}_k$
  - $\bar{V}$: the scaled average outperformance of the best model; a large $\bar{V}$ is evidence against H_0

Case study: the Reality Check of White (2000) (2/2)

- Null distribution by the stationary bootstrap of the whole vector f_t (keeps the correlation across rules and over time):
  - $\bar{V}_b^* = \max_k \sqrt{n}\,(\bar{f}_{k,b}^* - \bar{f}_k), \quad p = B^{-1}\sum_b 1\{\bar{V}_b^* > \bar{V}\}$
  - $b = 1, \ldots, B$: the resamples; $\bar{f}_{k,b}^*$: the mean of rule k in resample b; $1\{\cdot\}$: 1 if the condition holds, 0 otherwise
  - p: the share of resamples with a larger maximum than the observed one, i.e. a p-value for the whole search
- Sullivan, Timmermann and White (1999) apply it to 7846 technical rules on the Dow Jones; Hansen (2005) (SPA) and Romano and Wolf (2005) (StepM) refine it: Chapter 1
- Our application: 50 rules "long if the close is above its n-day moving average", $n = 5, 10, \ldots, 250$, on the BET
  - against buy-and-hold, no costs; mean block of 10 days

The Reality Check on the BET



- Bootstrap distribution of $\bar{V}^*$ under H_0 (999 stationary-bootstrap resamples) and the observed $\bar{V}$; two subsamples

ATS_ch0_data_snooping

Interpreting the Reality Check

- 2001–2012 ($n = 2\,993$ days): the best rule MA(5) beats buy-and-hold by 0.055% a day, $t = 2.51$, naive one-sided $p = 0.006$
  - Reality Check $p = 0.06$: not significant at 5% once the search over 50 rules is accounted for
- 2014–2026: the same rule, $t = 0.97$, Reality Check $p = 0.38$: whatever existed did not survive
- The short-MA winner fits the positive $\hat{\rho}_1$ of BET returns (stale prices)
  - transaction costs would eat the gain
- Pre-register the rule universe and the sample split before looking at the results (Stage 1 of the project)

Recap: multiple testing

- The p-value of the best of K is not the p-value of a pre-specified test
- FWER (Bonferroni, Holm) or FDR (Benjamini–Hochberg); for correlated forecasts: bootstrap the maximum (Reality Check)
- Forecast comparison tests (Diebold–Mariano, SPA, Model Confidence Set) are the subject of Chapter 1

Seeds and Monte Carlo experiments

- One seed per project, set once: `rng = np.random.default_rng(2026)`; pass `rng` to every function
  - Parallel runs: `np.random.SeedSequence(2026).spawn(k)` gives independent streams; never reuse a global state
- Report what makes a simulation checkable
  - DGP, T, number of replications R, burn-in, seed, Monte Carlo standard error
  - Results must not depend on the seed beyond the Monte Carlo error: rerun with a second seed
- Every chart and every number of this chapter is produced by one script with seed 2026 (Quantlets/Ch_00)

Versioned data

- Macro data are revised: the GDP you download today is not the GDP a forecaster saw in real time
  - Real-time vintages: Croushore and Stark (2001) (Philadelphia Fed), ALFRED for FRED series; Eurostat revises national accounts every quarter
- Save a dated snapshot of every series used, with the query that produced it
  - Record the download date, the dataset code (e.g. `namq_10_gdp`) and a checksum (`sha256`) of the file
  - The course market data are saved once and end on 18 September 2026: every student gets the same numbers
- A forecast evaluated on revised data answers a different question than the one asked in real time

Replication packages

- What economics journals now require Vilhuber (2020), Christensen and Miguel (2018)
  - a README with a data availability statement, the order of the scripts and the expected run time
  - code that regenerates every table and figure from raw data, without manual steps
  - the computational environment: `requirements.txt` with versions, or a Colab notebook
- For the ATS project (Stage 2)
  - a GitHub repository: `data/`, `code/`, `output/`, README, AI_USE.md, AI_ERRORS.md
  - one command or one notebook reproduces every number in the report
- Test: a colleague clones the repository on a clean machine and gets the same numbers

Pre-registration and the analysis plan

- Stage 1 of the project fixes, before any estimation:
  - the hypothesis, the sample and its split (estimation, evaluation), the benchmark
  - the inference method (HAC rule, bootstrap, critical values) and the multiple-testing correction
  - the list of specifications that will be reported, all of them
- Why: the garden of forking paths turns many small choices into an implicit search
  - Every deviation from the plan is allowed but reported, with its reason
- An AI assistant makes the search cheaper: AI_USE.md records the prompts that shaped a decision

Recap: reproducible research

- Seed once, pass the generator, report R and the Monte Carlo error
- Dated, checksummed data snapshots; real-time vintages when the question is about real-time forecasting
- A replication package that runs from raw data to every number; a plan fixed before the results

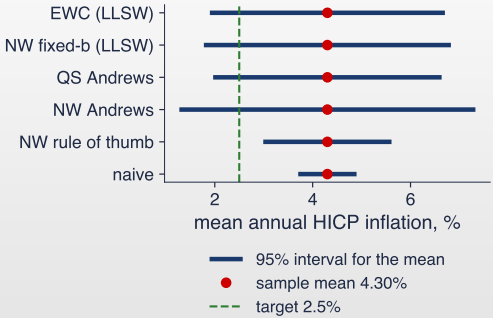
An open question: valid inference for a near-integrated mean

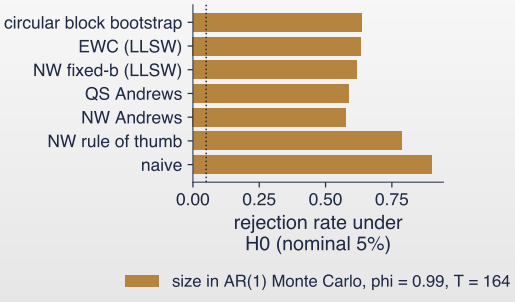
- Has Romanian inflation averaged the BNR target since 2013? Formally, $H_0 : \mu = 2.5$ for the HICP annual rate, 164 months to August 2026
  - $\hat{\rho}_1 = 0.99$: the series is close to a unit root, and overlapping by construction
- Open: no kernel, EWC or bootstrap rule controls size at this persistence (Monte Carlo above)
  - Müller (2014) and Lazarus, Lewis and Stock (2021) map the frontier; which test should a central-bank watcher use with 10–15 years of data?
- A question with policy content (credibility of the target) and a methodological core (HAR inference at $\phi \approx 1$)

The AI-assisted discovery loop

- **Literature**: `"List peer-reviewed tests for the mean of a highly persistent series since 2010, with DOI and the Monte Carlo design."`
  - Check: resolve every DOI on Crossref; read the size tables yourself (Semantic Scholar, Elicit for search)
- **Hypothesis and code**: `"Write Python for NW fixed-b and EWC tests of a mean, and a Monte Carlo of their size at phi = 0.98, T = 164."`
  - Check: reproduce a published size number first (e.g. the LLSW tables), then change one thing at a time
- **Robustness and critique**: `"Act as a hostile referee: why could this test of the inflation mean be misleading?"`
  - Expect: breaks in the mean (2013, 2022), the CPI–HICP gap, overlap, the choice of start date
- **The human checks**: formulas, timing of information (no look-ahead), every number against the code; the tools are generic (Claude, ChatGPT, Gemini, Copilot)

Mini-case: six tests of the mean HICP inflation





- Left: 95% intervals for the mean HICP inflation since 2013 (mean 4.30%); right: size of each test in a Monte Carlo with the fitted AR(1), $\phi = 0.99$, $T = 164$

ATS_ch0_ai_discovery

Interpreting the mini-case

- A typical assistant proposal: "NW with $\lfloor 4(T/100)^{2/9} \rfloor$ lags" gives $t = 2.69$ and rejects $\mu = 2.5$
  - Its actual size at this persistence: 79% instead of 5%; the naive test: 90%
- NW-Andrews ($t = 1.17$), fixed-b ($t = 1.59$, critical value 2.23) and EWC ($t = 1.63$, $\nu = 12$) do not reject
  - But their sizes are still 58%–63%: a non-rejection is not evidence for the target either
- Honest conclusion: with 14 years of a near-integrated series the data cannot settle the question; that is the open part

Project idea

- **Question**: which HAR test is reliable for the mean of Romanian and euro-area inflation, 2005–2026?
  - Replicate first: one size table of Lazarus et al. (2018) (their AR(1) design); explain every gap
- **Extension**
  - add the tests of Müller (2014); calibrate the Monte Carlo to HICP data of all EU countries (Eurostat)
  - report size and power on one frontier per country; test the target hypothesis with the test that controls size
- **Deliverables**: a dated pre-analysis plan; a replication package; AI_USE.md with the prompts that shaped decisions; AI_ERRORS.md with each invented reference or wrong number caught

Key takeaways

- The precision of a mean or a slope is governed by the long-run variance $\Omega = 2\pi f(0)$, not by γ_0
- Ergodicity gives consistency; MDS or mixing structure gives the CLT; the HAC estimator gives the standard error
- Kernel, bandwidth and critical values are one decision: NW with $1.3\sqrt{T}$ and fixed-b, or EWC, by default
- Block bootstraps resample dependence; the wild bootstrap does not
- Near unit roots defeat all of them: measure the size of your test by Monte Carlo at your persistence
- Searches need search-adjusted p-values; projects need replication packages and a plan fixed in advance

Self-assessment

- What is the long-run variance of an MA(1) with $\theta = -0.5$ and $\sigma^2 = 1$?
- Why can the truncated kernel give a negative variance while the Bartlett kernel cannot?
- Which standard error would you use for the mean of daily S&P 500 returns?
- Which standard error would you use for the mean of their squares?
- Why is the wild bootstrap invalid for a regression with overlapping four-quarter growth?
- You tried 40 specifications and report the best one: which p-value should you report?

Next chapter and further reading

- **Next**: Chapter 1, forecast evaluation, scoring rules and combination
  - Diebold–Mariano is a HAC t-test on loss differentials; SPA and the Model Confidence Set use the stationary bootstrap of this chapter
- **Further reading**
  - Theory: Hamilton (1994) (Ch. 7 and 10), Bradley (2005), Lahiri (2003)
  - Practice: Lazarus et al. (2018), Lazarus, Lewis and Stock (2021), Müller (2014)
  - Snooping: White (2000), Hansen (2005), Romano and Wolf (2005), Harvey, Liu and Zhu (2016)
- **Code**
  - Quantlets: github.com/danpele/Advanced-Time-Series/tree/main/Quantlets/Ch_00
  - Python: `statsmodels` (`cov_type="HAC"`), `arch.bootstrap` (`StationaryBootstrap`, `optimal_block_length`, `SPA`)

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